

GO Enrichment using goseq

Overview

In this lab, we'll look at how to use an R package called goseq to identify enriched gene ontology (GO) terms. For this analysis, we'll be using the differential analysis results we generated using DESeq.

(Why are we using the list of differentially expressed genes produced by DESeq and not cuffdiff?)

Introduction

goseq is an R package that provides functions to look for **enriched gene ontology terms (GO)** in our differentially expressed genes.

WHAT ARE GO TERMS?

GO terms provide a standardized vocabulary to describe genes and gene products from different species. GO terms allow us to assign functionality to genes. The following properties are described for gene products:

- **cellular component**, describes where in a cell a gene acts, what cellular unit the gene is part of
- **molecular function**, describes the function carried out by the gene, such as binding or catalysis;
- **biological process**, a set of molecular functions, with a defined beginning and end, makes up a biological process. This describes biological phenomenon like DNA replication.

All GO terms have an ID that looks like GO:0006260.

All GO terms have a list of genes that belong to that particular term.

WHAT IS GO ENRICHMENT?

For GO enrichment, we take the following things into account:

- A. Total number of genes we are looking at.
- B. Number of genes of interest, that is, in our DEG list.
- C. Total number of genes in the GO term
- D. Number of genes from our genes of interest that are also in the GO term.

If the number of genes from our list that belong to GO term GO:0001 (D) is significant compared to the total number of genes in that GO term (C) and the total number of genes in our experiment (A), we consider that GO term to be enriched in our data.

WHAT DOES THIS MEAN TO US?

Many genes may be changing, but they may all be linked to similar biological processes. From a list of changing genes → list of affected biological processes. We can better elucidate the biological events that are represented by our differential gene finding.

We also reduce the dataset considerably- from large number of genes to a smaller number of functions/processes.

We go from up and downregulated genes between two conditions to up and down regulated processes between two conditions.

INPUT TO GOSEQ

- A list of all genes tested.
- A list of just the genes of interest, in this case, significantly differentially expressed genes.

OTHER NOTES ABOUT GOSEQ

- It needs information about our genome, particularly length of genes. This is available in a particular bioconductor package for many model genomes.
- If your genome is not one of the genomes supported by that package, you can attempt to create the required files about your genome using commands mentioned in the goseq manual.
- goseq can also be used to identify enriched pathways, like KEGG pathways

What are your other options?

- If you are trying to find enriched GO terms using a list of novel transcripts, BLAST2GO, AmiGO
- Web services like [Enrichr](#), DAVID
- For pathway analysis: kegg, Ingenuity pathway analysis

Get set up

Get set up

```
1 Make sure you are in the right location
2
3 cd $SCRATCH/my_inaseq_course/day_3/cuffdiff_results
```

A) First load R and enter R environment

```
1 #module load R will not work for this R package because its needs the latest version of R. So, we'll load a local
  version.
2
3 /work/01184/daras/bin/R
4
```

B) **DONT RUN THIS- THESE HAVE ALREADY BEEN INSTALLED** Within R environment, if you needed to install packages

```
1 #DONT RUN ANY OF THIS RIGHT NOW FOR THE CLASS
2
3 source("http://bioconductor.org/biocLite.R")
4 biocLite("goseq")
5
6 #package to pull out annotated information about our genome and genes
7 biocLite("geneLenDataBase")
8
9 #package to load the GO terms specific to drosophila
10 biocLite("org.Dm.eg.db")
```

C) While that is installing, let's look at how your input files need to look.

FROM DESeq2 output:

```
","id","baseMean","baseMeanA","baseMeanB","foldChange","log2FoldChange","pval","padj"
```

```
"131","FBgn0000370",7637.91654540105,4217.77033402576,11058.0627567763,2.62177925326286,1.39054621964  
443,1.2887282997047e-116,7.22484613473489e-113
```

```
"2489","FBgn0025682",6038.35042952997,3300.21617337019,8776.48468568974,2.65936660649935,1.41108267  
336748,1.36704751839828e-116,7.22484613473489e-113
```

.....

INPUT FILE 1: DEG (contains the 76 genes that meet our fold change and p value cut offs)

FBgn0000370

FBgn0025682

FBgn0086904

.....

INPUT FILE 2: ALL (contains all 14869 genes)

FBgn0000370

FBgn0025682

FBgn0086904

.....

C) Load all the libraries.

```
1 library("goseq")  
2 library("geneLenDataBase")  
3 library("org.Dm.eg.db")
```

D) Read in our two input files

```
1 DEG<-read.table("DEG", header = FALSE)  
2 ALL<-read.table("ALL", header = FALSE)
```

E) Convert the DEG and ALL data objects to vectors

```
1 class(DEG)  
2 DEG.vector <- c(t(DEG))  
3 ALL.vector<-c(t(ALL))
```

Data frame:

FBg..		
FBg...		
FBg...		

Transpose:

FBg...	FBg...	FBg...
--------	--------	--------

Vector:

FBg..., FBg..., FBg...

F) Construct a new vector that adds a 0 next to every gene that is not in our DEG list and a 1 next to every gene that is in our DEG list.

```
1 gene.vector=as.integer(ALL.vector%in%DEG.vector)
2 names(gene.vector)=ALL.vector
3 #lets explore this new vector a bit
4 head(gene.vector)
5 tail(gene.vector)
```

G) Weigh the gene vector by lengths of our genes. This length information is pulled from a different package and we need to specifying the exact name of our genome and gene IDs in that package.

```
1 pwf=nullp(gene.vector,"dm3","ensGene")
```

F) Find enriched GO terms

```
1 GO.wall=goseq(pwf,"dm3","ensGene")
2
3 #How many enriched GO terms do we have
4 class(GO.wall)
5 head(GO.wall)
6 nrow(GO.wall)
```

G) Find only enriched GO terms that are statistically significant

```
1 enriched.GO=GO.wall$category[GO.wall$over_represented_pvalue<.05]
2 #NOTE: They recommend using a more stringent multiple testing corrected p value here
3
4 #How many GO terms do we have now?
5 class(enriched.GO)
6 head(enriched.GO)
7 length(enriched.GO)
```

H) But these are just IDs - Get more biologically meaningful results

```
1 library(GO.db)
2 capture.output(for(go in enriched.GO[1:258]) { print(GOTERM[[go]])
3 cat("-----\n")
4 }
5 , file="SigGo.txt")
```

I) OUTPUT

```
1 less SigGo.txt
```

File is

<http://web.corral.tacc.utexas.edu/BioITeam/>

SigGo.txt

GOID: GO:0030336

Term: negative regulation of cell migration

Ontology: BP

Definition: Any process that stops, prevents, or reduces the frequency,
rate or extent of cell migration.

Synonym: down regulation of cell migration

Synonym: down-regulation of cell migration

Synonym: downregulation of cell migration

Synonym: inhibition of cell migration

GOID: GO:0040013

Term: negative regulation of locomotion

Ontology: BP

Definition: Any process that stops, prevents, or reduces the frequency,
rate or extent of locomotion of a cell or organism.

Synonym: down regulation of locomotion

Synonym: down-regulation of locomotion

Synonym: downregulation of locomotion

Synonym: inhibition of locomotion
